

**USER INFORMATION  
READ CAREFULLY**

**INFUSOL® QSD1  
SODIUM CHLORIDE 0.18% w/v & DEXTROSE 4.23% w/v INTRAVENOUS INFUSION BP**

**Composition:**

|                       |          |
|-----------------------|----------|
| Sodium Chloride BP    | 1.8 g/L  |
| Dextrose Anhydrous BP | 42.3 g/L |

|                                 |                                |              |
|---------------------------------|--------------------------------|--------------|
| <b>Electrolytes (approx) :</b>  | <b>mmol/L</b>                  | <b>mEq/L</b> |
| Sodium Ion (Na <sup>+</sup> )   | 30.8                           | 30.8         |
| Chloride Ion (Cl <sup>-</sup> ) | 30.8                           | 30.8         |
| Osmolarity                      | : 296 mOsm/L                   |              |
| Caloric Value                   | : Approx. 710kJ/L = 170 kcal/L |              |

**Product Description:**

A colourless or faintly strawed coloured solution.

0.18% Sodium Chloride and 4.23% Dextrose Intravenous Infusion BP represent a slightly hypertonic IV solution. The outstanding feature of this solution is the markedly low electrolyte content which makes it particularly suitable for use in case of hypertonic dehydration. Since the young are known to require less electrolytes than adults, this solution is ideally suited for use in paediatrics.

**Pharmacodynamic:**

Glucose with Sodium Chloride supplies energy, along with correction of electrolyte imbalance. It is also supplies water along with essential salts. This is because Sodium Chloride is distributed easily and builds up the salt level rapidly. Thus the second and third step of the dehydration therapy is achieved. This combination of Sodium Chloride with Glucose eliminates the problem of incompatibility which occurs when plain Glucose is administered with preserved blood. This results in aggregation of erythrocytes which may lead to transfusion problems.

The Sodium content and the liquid metabolism of the body are closely coupled to each other. Each deviation of the plasma sodium concentration from the physiological one simultaneously affects the fluid status of the body.

An increase in the Sodium content of the body also results in a reduction of the body's free water content independent of the serum osmolality. Glucose is metabolised ubiquitously as the natural substrate of the cells of the body. Under physiological conditions Glucose is the most important energy-supplying carbohydrate with a caloric value of ca. 16 kJ or 3.75 kcal/g. Nervous tissue, erythrocytes and medulla of the kidneys are amongst the tissues with an obligate requirement for Glucose. In adults, the concentration of Glucose in the blood is 70 – 100 mg/100 mL, or 3.9 – 5.6 mmol/L (fasting).

Glucose serves for the synthesis of glycogen as the storage form of carbohydrates and it is subject to glycolysis to pyruvate and lactate for energy production in the cells. Glucose also serves to maintain the blood sugar level and for the synthesis of important body components. It is primarily insulin, glucagon, glucocorticoids and catecholamines that are involved in the regulation of the blood sugar concentration.

A normal electrolyte and acid-base status is a prerequisite for the optimal utilisation of administered glucose. So, an acidosis in particular can indicate impairment of the oxidative glucose metabolism.

**Pharmacokinetics:**

**Absorption**

As the solution is administered by intravenous infusion the bioavailability of the solution is 100%.

**Distribution**

The total Sodium content of the body is ca. 80 mmol/kg of which ca. 97% is extracellular and ca. 3% intracellular. The daily turnover is ca. 100 - 180 mmol (corresponding to 1.5 - 2.5 mmol/kg body weight). On infusion Glucose is first distributed in the intravascular space and then is taken up into the intracellular space. The total body Chloride in adults is about 33 mmol/kg body weight. Serum Chloride is maintained at 98 - 108 mmol/L.

**Biotransformation**

The kidneys are the major regulator of the Sodium and Water balances. In co-operation with the hormonal control mechanisms (renin-angiotensinaldosterone system, antidiuretic hormone) and the hypothetical natriuretic hormone they are primarily responsible for keeping the volume of the extracellular space constant and regulating its fluid composition.

In glycolysis glucose is metabolised to pyruvate or to lactate. Lactate can be partially re-introduced into the Glucose metabolism (Cori cycle). Under aerobic conditions pyruvate is completely oxidized to Carbon Dioxide and Water. Glucose utilisation disturbances (Glucose intolerance) can occur under conditions of pathological metabolism. These mainly include diabetes mellitus and states of metabolic stress (e.g., intra-, and postoperatively, severe disease, injury), hormonally mediated depression of Glucose tolerance, which can even lead to hyperglycaemia without exogenous supply of the substrate. Hyperglycaemia can - depending on its severity - lead to osmotically mediated renal fluid losses with consecutive hypertonic dehydration, to hyperosmotic disorders up to and including hyperosmotic coma.

Metabolism of glucose and electrolytes are closely related to each other. Insulin facilitates potassium influx into cells. Phosphate and Magnesium are involved in the enzymatic reactions associated with Glucose utilization. Potassium, Phosphate and Magnesium requirements may therefore increase following Glucose administration and may therefore have to be monitored and supplemented according to individual needs. Especially cardiac and neurological functions may be impaired without supplementation.

**Elimination**

Sodium and Chloride are excreted via sweat, urine and the gastrointestinal tract. Chloride is exchanged for Hydrogen Carbonate in the tubule system and is, thus, involved in the regulation of the acid base balance. The final products of the complete oxidation of Glucose are eliminated via the lungs (Carbon Dioxide) and the kidneys (water). Practically no Glucose is excreted renally by healthy persons. In pathological metabolic conditions (e.g., diabetes mellitus, post-aggression metabolism) associated with hyperglycaemia (blood glucose concentrations of more than 120 mg/100 ml or 6.7 mmol/l), Glucose is also excreted via the kidneys (glucosuria) when the maximum tubular resorption capacity (180 mg/100 ml or 10 mmol/l) is exceeded.

**Indication:**

- Hypertonic (esp. hypernatraemic) dehydration
- Supply of free water for electrolyte maintenance in paediatrics
- Caloric supply
- Vehicle solution for supplementary medication.

**Recommended Dosage:**

The dosage depends on the age, weight, clinical and biological (acid-base balance) conditions of the patient, concomitant therapy and should be determined by the consulting physician.

Infants : 100 to 1000mL/day  
Children : 200 to 2000mL/day  
Adults : 1 to 3L/day

**Drop rate :** Up to 120 drops/min corresponding to 360 - 540 mL/h.

**Route of Administration: Intravenous (IV)**

**Contraindications:** Hyperhydration, Oedema, Hypotonic dehydration, Cardiac insufficiency

**Paediatric population**

This product should not be used in children except in paediatric specialist settings (such as renal, hepatic and cardiac units, high dependency units and intensive care units) under expert medical supervision.

**Warning and Precautions:**

Intravenous infusions of concentrated dextrose injection are liable to cause local vein irritation. Hypophosphatemia may occur after prolonged use.

INFUSOL® QSD1 should not be used for fluid substitution, especially rehydration therapy, without adequate electrolyte administration, because this could lead to markedly decreased serum electrolyte values, notably severe hyponatraemia, with potentially detrimental effects on the patient, e.g., brain damage or heart affections. Especially children, elderly patients and patients in poor general condition are at risk.

INFUSOL® QSD1 should not be used for routine rehydration or fluid maintenance therapy. In patients with hyponatraemia or hypokalaemia administration of this solution must be accompanied by additional substitution of Sodium or Potassium as necessary to normalise the serum levels of these electrolytes.

In diabetic patients, the amount of infused Glucose has to be taken into account and insulin requirements may be modified. In patients with chronic hypernatraemia, the serum Sodium concentration should not be lowered faster than at a rate of 0.5 mmol/L/h. Administration of Glucose solutions is not recommended after acute ischaemic strokes as hyperglycaemia was reported to worsen ischaemic brain damage and impair recovery.

Clinical monitoring should include checks of the serum electrolytes (especially Potassium), Glucose level, the acid-base and water balance. If the oxidative metabolism of Glucose is impaired close monitoring of blood glucose levels is recommended in order to prevent hyperglycaemia.

*Please note: If this solution is used as vehicle solution the safety information of the additive provided by the respective manufacturer has to be taken into account. The solution should not be administered through the same infusion equipment simultaneously, before or after an administration of blood because of the possibility of pseudo-agglutination.*

#### Paediatric population

Intravenous fluid therapy should be closely monitored in the paediatric population as they may have impaired ability to regulate fluids and electrolytes. Adequate urine flow must be ensured and careful monitoring of fluid balance, plasma and urinary electrolyte concentrations are essential.

The infusion of hypotonic fluids together with the non-osmotic secretion of ADH (in pain, anxiety, the post-operative state, nausea, vomiting, pyrexia, sepsis, reduced circulating volume, respiratory disorders, CNS infections, and metabolic and endocrine disorders) may result in hyponatraemia. Hyponatraemia can lead to headache, nausea, seizures, lethargy, coma, cerebral oedema and death, therefore acute symptomatic hyponatraemic encephalopathy is considered a medical emergency.

#### Cautions:

The compatibility of any additives to this solution should be checked before use. Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

#### Interactions with Other Medicaments:

The compatibility of any additives to this solution should be checked before use. Corticosteroids are associated with the retention of Sodium and Water.

#### Pregnancy and Lactation:

##### Pregnancy

There is a limited amount of data from the use of Sodium Chloride and Glucose in pregnant women. Caution should be exercised when prescribing to pregnant women especially in the presence of pre-eclampsia. Careful monitoring of blood glucose is necessary.

##### Breast-feeding

It is known that Glucose/metabolites are excreted in human milk, but at therapeutic doses of INFUSOL® QSD1 has no effects on the breastfed newborns/infants are anticipated.

#### Side Effects:

Intravenous infusions of Concentrated Dextrose Injection are liable to cause local vein irritation. Hypophosphatemia may occur after prolonged use.

#### Symptoms and Treatment of Overdose :

Dextrose solutions IV can cause fluid overload resulting in dilution of serum electrolyte concentration, congested state or pulmonary oedema. Significant hyperglycaemia, hyperosmolar syndrome and glucosuria may occur with too rapid administration of hypertonic solutions.

Hyperglycaemia and glucosuria may occur if rate of administration is too fast or in case of metabolic insufficiency. To minimise these conditions, slow infusion rate, monitor blood and urine Glucose; if necessary, stop administration of Dextrose or/and administer insulin.

#### Incompatibilities:

When mixing with other medicinal products possible incompatibilities should be considered. It should be remembered that the solution has an acid pH, which can cause precipitation in the mixture.

#### Shelf Life:

5 years from manufacturing date in the proposed storage condition for LDPE Bottle. Do not use after expiry.

#### Storage Conditions:

Do not store above 30°C.

#### Dosage form and packaging available:

500mL x 10 LDPE plastic bottle per carton

500mL x 20 LDPE plastic bottle per carton

#### Manufacturer / Product Registration Holder:



**AIN MEDICARE SDN.BHD.**

Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

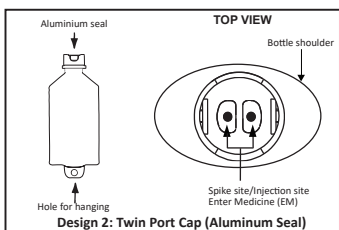
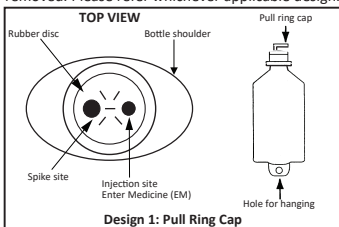
#### HANDLING INSTRUCTION

#### AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE

The bottle is made from injectable grade polyethylene.

#### PARTS OF THE PLASTIC BOTTLE

Top view after pull ring cap and aluminum seal is removed. Please refer whichever applicable design.



#### INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

